

Modes of Heat Transfer :

Conduction	Convection	Radiation
Need medium	Need medium	No medium
No particle transfer	Particle transfer	Energy transfer through EM waves
atoms vibrate		

Thermal Conduction :

→ In steady state conduction, the rate at which heat flows from one end to another is constant.

i.e. $\frac{dQ}{dt} = \text{const.}$
 → Called 'Thermal Current'?



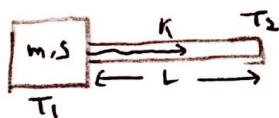
$$i_{Th} = \frac{dQ}{dt} = \frac{KA}{L} (T_1 - T_2) \quad (\text{J/s})$$

K → Thermal conductivity (W/m-K) of material

A → Cross section of Rod

L → Length of rod.

→ Suppose heat is being drawn from a hot body at temp T_1 , then rate of cooling is given by

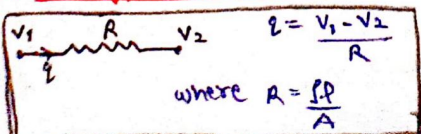


$$\frac{dT}{dt} = - \frac{KA(T_1 - T_2)}{mSL}$$

→ If $\frac{dT}{dt} = +ve$ ⇒ rise in temp of body

→ If $\frac{dT}{dt} = -ve$ ⇒ Fall in temp. of body

Analogy with Ohm's law :



Similarly,

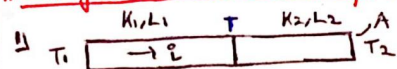
$$i = \frac{T_1 - T_2}{L/KA} = \frac{T_1 - T_2}{R_{Th}}$$

R_{Th} → Thermal Resistance

$$[R_{Th} = \frac{L}{KA}]$$

NOTE: All concepts of series & parallel combination also valid.

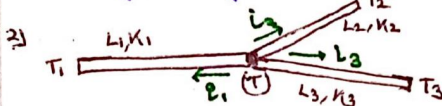
Interface and Junction Temp :



Solⁿ i will be same in both rods,

$$\frac{T_1 - T}{L_1/K_1A} = \frac{T - T_2}{L_2/K_2A}$$

→ Can find T from this eqⁿ.

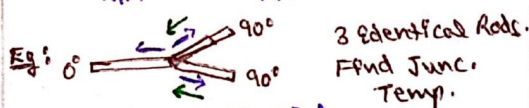


Solⁿ $i_1 + i_2 + i_3 = 0$

$$\Rightarrow \frac{T - T_1}{R_1} + \frac{T - T_2}{R_2} + \frac{T - T_3}{R_3} = 0$$

where

$$R_1 = \frac{L_1}{K_1A}, R_2 = \frac{L_2}{K_2A}, R_3 = \frac{L_3}{K_3A}$$



Since rods are identical,

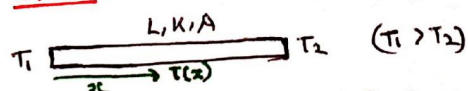
$$R_1 = R_2 = R_3$$

$$\frac{T - T_1}{R} + \frac{T - T_2}{R} + \frac{T - T_3}{R} = 0$$

$$3T - 180 = 0 \Rightarrow T = 60^\circ$$

We can't comment on correct dirⁿ of current, only after finding junction Temp.

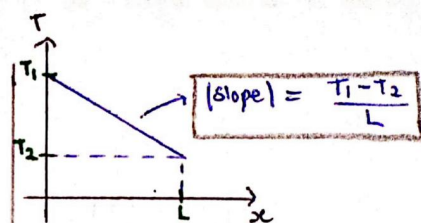
Temperature variation at steady state :



At steady state, uniform i flows throughout the body. So

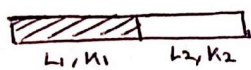
$$\frac{T_1 - T(x)}{x/KA} = \frac{T_1 - T_2}{L/KA}$$

$$\Rightarrow T(x) = - \left(\frac{T_1 - T_2}{L} \right) x + T_1$$



Equivalent Thermal Conductivity :

1) Series Combination :

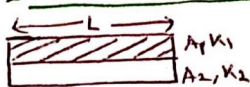


$$R_{eq} = R_1 + R_2$$

$$\frac{L_1 + L_2}{K_{eq}A} = \frac{L_1}{K_1A} + \frac{L_2}{K_2A}$$

$$\Rightarrow K_{eq} = \frac{K_1 K_2 (L_1 + L_2)}{L_1 K_2 + L_2 K_1}$$

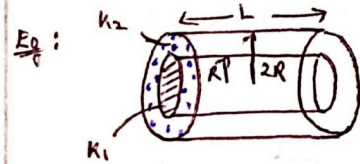
2) Parallel Combination :



$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$K_{eq} \frac{(A_1 + A_2)}{L} = \frac{K_1 A_1}{L} + \frac{K_2 A_2}{L}$$

$$K_{eq} = \frac{K_1 A_1 + K_2 A_2}{A_1 + A_2}$$



Find K_{eq} if no heat loss.

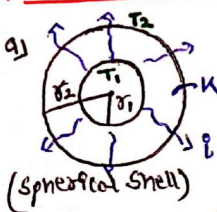
$$\text{Solⁿ } \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\Rightarrow K_{eq} \frac{(\pi R^2)}{L} = \frac{K_1 \pi R^2}{L} + \frac{K_2 \pi (R^2 - R_1^2)}{L}$$

$$4K_{eq} = K_1 + 3K_2$$

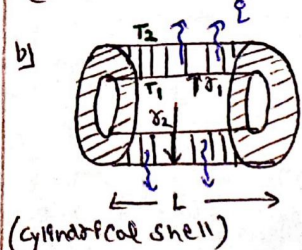
$$\Rightarrow K_{eq} = \frac{K_1 + 3K_2}{4}$$

Radial Heat Conduction :



$$i = \frac{T_1 - T_2}{R_{Th}}$$

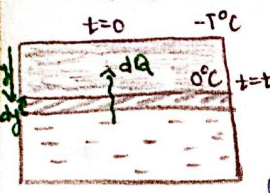
$$\text{where } R_{Th} = \frac{r_2 - r_1}{4\pi K r_1 r_2}$$



$$R = \frac{T_1 - T_2}{R_{Th}}$$

$$\text{where } R_{Th} = \frac{\ln(r_2/r_1)}{2\pi K L}$$

Freezing of Lake (Time taken to freeze depth y)



At $t=0$, lake starts to freeze from top.

(ii) At time $t=t$, y depth freezes

(iii) In next dt time, dy depth more freezes by losing dQ heat.

So, $\frac{dQ}{dt} = \frac{KA}{y} \cdot T \Rightarrow dm \cdot L_f = \frac{KA \cdot T \cdot dt}{y}$

$\Rightarrow \rho A dy \cdot L_f = \frac{KA}{y} T \cdot dt$

$\Rightarrow \frac{\rho L_f}{KT} \int y \cdot dy = \int dt$

$\Rightarrow t = \frac{\rho L_f}{2KT} \cdot y^2$

Here, $\rho \rightarrow$ Density of liquid water
 $L_f \rightarrow$ Latent heat of fusion
 $K \rightarrow$ Thermal conductivity

Radiation :

$\rightarrow$ Radiation means emission of EM waves.

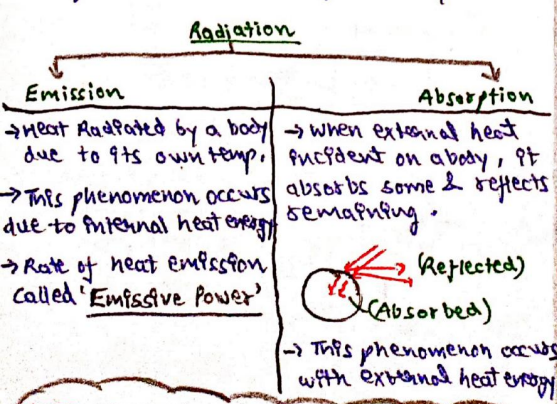


$\rightarrow$ There are following 3 theories related to Radiation.

1) Prevost Theory of Heat Exchange :

Any body at temperature greater than zero Kelvin emits/radiates energy out of it in the form of EM waves continuously.

At room temp., usually IR or lower energy rays are emitted by the body.

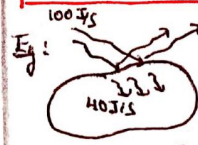


NOTE: Reflection of heat energy is not emission. Both are different.

Absorptivity (a) & Reflectivity (r) :-

$a = \frac{\text{Total Energy absorbed}}{\text{Total Energy Incident}}$ $[a < 1]$

$r = \frac{\text{Total Energy reflected}}{\text{Total Energy Incident}}$ $[r < 1]$



$a = \frac{40}{100} = 0.4$

$r = \frac{60}{100} = 0.6$

$[a+r = 0.4+0.6 = 1]$

Ideal Black Body :-

$\rightarrow$ A hypothetical body that can absorb all the heat energy incident on it and does not reflect at all

$[e = a = 1]$

Ferry's Black Body Model :-

$\rightarrow$ Consists of a double walled hollow sphere with a fine hole and a pointed projection in front of the hole.

$\rightarrow$ Inner walls are covered with lamp black, & the model is able to absorb the light radiation entering it, completely.



Relative Emissivity (e) :-

Any body, which is not a black body, is called a 'Grey Body'

$e = \frac{\text{Heat released by Grey body}}{\text{Heat released by Black body}} \quad (e < 1)$

NOTE: For calculation of e

(i) Dimensions of both Grey & Black body should be same

(ii) Both bodies should be at same temp. T

2) Kirchhoff's Law :-

$\rightarrow$ Good emitters are also good absorbers i.e. $[e = a < 1]$ (For Grey Bodies)

$\rightarrow$ Thus, Black body is best emitter in the universe (as $e=1$)

3) Stefan's Law :-

$\rightarrow$ This law enables us to calculate rate of heat emission by Black Bodies.

$\rightarrow$ For a black body,

$\frac{dQ}{dt} \propto A \propto T_b^4 \Rightarrow \frac{dQ}{dt} = \sigma \cdot A \cdot T_b^4$

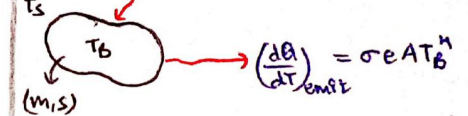
where, $\sigma \rightarrow$ Stefan Boltzmann Constant $(5.67 \times 10^{-8} \text{ ST unit})$
 $A \rightarrow$ Surface Area
 $T_b \rightarrow$ Temp. of Body

$\rightarrow$ This law was further modified for all type of bodies as follows.

$\frac{dQ}{dt} = \sigma e A T_b^4$ $\leftarrow$ Called Stefan-Boltzmann Law

(e $\rightarrow$ Emissivity of the body)

$\frac{(dQ)}{dt}_{\text{absorb}} = \sigma e A T_s^4 = \sigma a A T_b^4$



(where, $T_s \rightarrow$ Surround Temperature)

Net Rate of Heat Loss $\Rightarrow$ (Assuming $T_b > T_s$)

$\frac{dQ}{dt} = \sigma e A (T_b^4 - T_s^4)$

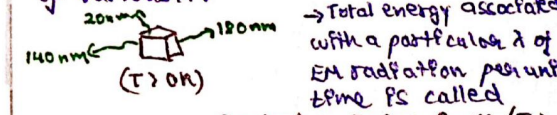
Net Rate of Cooling $\Rightarrow$ (-ve bcz T is using)

$\therefore \Delta Q = m \cdot S \cdot \Delta T \Rightarrow \frac{dQ}{dt} = -m \cdot S \cdot \frac{dT}{dt}$

$\Rightarrow \frac{dT}{dt} = -\frac{\sigma e A}{mS} (T_b^4 - T_s^4)$

Wien's Displacement Law :

$\rightarrow$ When a body above zero Kelvin radiates energy, it does so by radiation EM radiations of various λ .

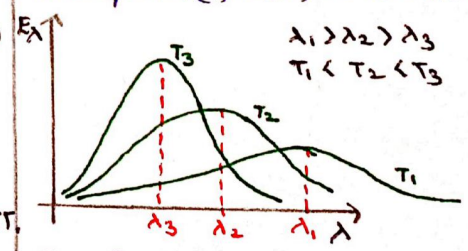


$\rightarrow$ Total energy associated with a particular λ of EM radiation per unit time is called

Spectral Emissive Power (E_λ)

$\rightarrow \lambda$ corresponding to which (E_λ) is max. is (λ_m)

$\rightarrow$ We plot (λ) vs (E_λ) curve & observed



For a Particular body

$\lambda_m \cdot T = b$ ($b \rightarrow$ Wien's Constant) $= 2.8 \times 10^{-3} \text{ m} \cdot \text{K}$

$\rightarrow$ As $T \uparrow \Rightarrow \lambda_m \downarrow \Rightarrow$ Energy Radiated $\uparrow$

$\rightarrow$ Area under curve gives total emissive power of body at particular temp.

$\int_0^\infty E_\lambda \cdot d\lambda = E = \sigma \cdot T^4$

$\rightarrow$ Ratio of Area under the curve at two diff. temp. will be

$\frac{A_1}{A_2} = \left(\frac{T_1}{T_2}\right)^4$

$\rightarrow$ Maximum spectral emissive power is directly proportional to T^5 .

i.e. $(E_\lambda)_{\text{max}} \propto T^5$